

1. Administrative & Study Identification

Full Study Title: Study of Talazoparib With Enzalutamide in Men With DDR Gene Mutated mCSPC **Short Title/Acronym:** TALAPRO-3
Protocol Number: TALAPRO-3 (Pfizer Internal ID) **NCT Number:** NCT04821622 **Sponsor Name:** Pfizer Inc. **Investigational Product:** Talazoparib (Talzenna) in combination with Enzalutamide (Xtandi)
Phase of Development: Phase III **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether the addition of talazoparib to enzalutamide provides a statistically significant and clinically meaningful improvement in radiological progression-free survival compared with placebo plus enzalutamide in participants with DDR-deficient metastatic castration-sensitive prostate cancer (mCSPC). **Secondary Objectives:** To evaluate overall survival, objective response in measurable soft tissue disease, and duration of response in measurable soft tissue disease.

2.2 Background & Rationale Alterations in DNA damage repair (DDR) genes, such as homologous recombination repair (HRR) genes, are found in approximately 25% of metastatic prostate cancers. These alterations are associated with a worse prognosis and are less responsive to current standards of care, representing a group with a high unmet need. Talazoparib plus enzalutamide is already a standard of care in HRR gene-mutated metastatic castration-resistant prostate cancer (mCRPC). The TALAPRO-3 trial is exploring the safety and efficacy of the combination earlier in the disease course (in the mCSPC setting) to delay disease progression and extend the time patients can live without their cancer worsening.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Talazoparib/Placebo are blinded; Enzalutamide is open-label) **Randomization:** 1:1

3.2 Planned Enrollment & Duration Target **SN\$:** 599 randomized patients (approximately 550 men with mCSPC randomized) **Number of Sites:** Approximately 285 sites in 27 countries globally **Estimated Study Start:** March 2021 **Estimated Primary Completion:** March 2026

(Topline results demonstrating target endpoints met reported March 2026) **Trial Status Description:** Active but not currently recruiting

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male participants ≥ 18 years of age at screening (20 years for Japan, 19 years for Republic of Korea). 2. Histologically or cytologically confirmed adenocarcinoma of the prostate without neuroendocrine differentiation, small cell, or signet cell features. 3. Confirmation of DDR gene mutation status by prospective or historical analysis of blood (liquid biopsy) and/or de novo or archival tumor tissue using FoundationOne Liquid CDx or FoundationOne CDx. 4. Ongoing ADT with a GnRH agonist or antagonist must be initiated before randomization and continue throughout the study.

4.2 Exclusion Criteria 1. Prior treatment of mCSPC with docetaxel. 2. Disease spread limited strictly to regional pelvic lymph nodes, or presence of known/suspected brain metastases or active leptomeningeal disease. 3. A finding of a "superscan" (extensive diffuse bone metastasis) at baseline.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Investigational Product: Talazoparib (ATC Code: L01XK04) **Dosage/Frequency:** Talazoparib 0.5 mg once daily (or 0.35 mg once daily for participants with moderate renal impairment, eGFR 30-59 mL/min/1.73 m²) plus Enzalutamide 160 mg once daily. **Route of Administration:** Oral capsules **Duration of Treatment:** Continuous until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Ongoing androgen deprivation therapy (ADT). **Prohibited:** Prior systemic anti-cancer treatments for mCSPC (such as docetaxel) or other investigational agents.

6. Study Timeline & Visit Schedule

The study consists of 5 main periods: prescreening, screening, double-blind treatment, safety follow-up, and long-term follow-up.

Period	Window	Key Activities/Procedures	:---	:---
:---	Prescreening/Screening	Day -28 to 0	Informed Consent, Biomarker testing for DDR/HRR gene alterations (FoundationOne CDx), Baseline CT/MRI and Bone Scans	Double-Blind Treatment
		Approx. Every 4 Weeks		Randomization, First

Dose, Physical exams, Blood tests, Dispense IP, Safety & AE assessments | | **Safety Follow-Up** | ~28 days post-treatment| Standard adverse event collection and safety evaluations after last dose | | **Long-Term Follow-Up**| Every 8 to 12 Weeks | Radiographic Imaging (CT/MRI/Bone Scan) to assess disease progression and survival |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Radiological Progression-Free Survival. **Measurement Method:** Defined as the time from the date of randomization to first objective evidence of radiographic progression or death, whichever occurs first.

7.2 Secondary Outcomes * Overall Survival * Objective response in measurable soft tissue disease * Duration of response in measurable soft tissue disease

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard AE collection throughout the treatment phase and a dedicated safety follow-up approximately 28 days after the last dose. **Serious Adverse Events (SAEs):** Evaluated strictly; the observed safety profile of the combination was consistent with the known safety profiles of the individual agents, with no new safety signals identified. **Data Monitoring Committee (DMC):** External and independent review monitoring mechanisms utilized by the sponsor.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy assessment. **Sample Size Justification:** The trial was powered to detect a pre-specified target hazard ratio for rPFS based on an estimated 550 to 599 enrolled participants. **Handling of Missing Data:** Standard survival analysis methodologies and censoring applied for rPFS. Patients were stratified by de novo mCSPC vs. relapsed mCSPC, high-volume vs. low-volume disease, and BRCA vs. non-BRCA mutational status.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Strict onsite and remote monitoring of health assessments, laboratory tests, and imaging. **Audit Trail:** Centralized data review, electronic capture of PROs (Patient-Reported Outcomes), and source data verification.

11. Study Identifiers & Governance

Posting ID: 285779833733143477 **Primary ID:** TALAPRO-3 **Registry Number:** NCT04821622 **Sponsor/Lead Organization:** Pfizer Inc. **Study Title:** Study of Talazoparib With Enzalutamide in Men With DDR Gene Mutated mCSPC **Official Regulatory Title:** TALAPRO-3: A PHASE 3, RANDOMIZED, DOUBLE-BLIND, STUDY OF TALAZOPARIB WITH ENZALUTAMIDE VERSUS PLACEBO WITH ENZALUTAMIDE IN MEN WITH DDR GENE MUTATED METASTATIC CASTRATION-SENSITIVE PROSTATE CANCER

12. Clinical Framework (mCSPC Oncology Specific)

Disease Area: Prostate Cancer **Primary Condition:** Cancer of the Prostate / Metastatic Castration-Sensitive Prostate Cancer (mCSPC) **Diagnosis Threshold:** Presence of ≥ 1 alteration in specified DDR/HRR genes **Medical Methodology:** PARP Inhibition combined with Androgen Receptor Signaling Inhibition **Optimization Target:** Maximal delay in Radiological progression-free survival (rPFS)

13. Study Procedures & Methodology

Management Pathway: Targeted oral systemic therapy alongside continuous androgen deprivation therapy (ADT). **Education Protocol (HCP):** Investigator meetings and site initiation training on RECIST v1.1 / PCWG3 imaging assessment criteria. **Education Protocol (Patient):** Informed consent detailing the rationale for genomic biomarker screening and blinded oral therapies. **Quality Audit Mechanism:** Investigator and central review of radiographic progression endpoints.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 3 years of average treatment; followed until death or study withdrawal. **Onsite Visit Cadence:** Approximately 25 clinic visits over the course of treatment. **Remote Monitoring Frequency:** Adverse event reporting and telemedicine follow-ups as required. **Checklist Review Interval:** Regular disease staging (CT/MRI and Bone scans) every 8 or 12 weeks.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	Neeraj Agarwal, MD, FASCO	_____	06-Apr-2026	Clinical Operations Lead	[Name]	_____	06-Apr-2026
Lead Statistician	[Name]	_____	06-Apr-2026				